

**The due date for submitting this assignment has passed.**

**Due on 2026-04-29, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

**You have last submitted on: 2026-04-29, 11:51 IST**

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

1) Consider the following function

**1 point**

$$f(x, y, z) = x^2z + y^3e^{xy} + \sin(xyz).$$

Which of the following option(s) is (are) true?

- ☐  $f_{xx} = 2z + y^5e^{xy} - yz \sin(xyz)$
- ☒  $f_{xx} = 2z + y^5e^{xy} - (yz)^2 \sin(xyz)$
- ☐  $f_{zy} = \cos(xyz)x^2yz$
- ☐  $f_{xz} = 2x - xy^2z \sin(xyz)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$f_{xx} = 2z + y^5e^{xy} - (yz)^2 \sin(xyz)$$

2) Number of critical points of the function  $f(x, y, z) = e^x(x^2 - y^2 - 2z^2)$  is

2

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

**1 point**

3) Choose the correct statement(s) about  $f(x, y) = \sqrt{x^2 + y^2}$ .

**1 point**

- ☐  $f$  has no critical points.
- ☒  $(0, 0)$  is the only critical point of  $f$ .
- ☐  $(0, 0)$  is a local maximum of  $f$ .
- ☒  $(0, 0)$  is a local minimum of  $f$ .
- ☒

$(0, 0)$  is a global minimum of  $f$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

$(0, 0)$  is the only critical point of  $f$ .

$(0, 0)$  is a local minimum of  $f$ .

$(0, 0)$  is a global minimum of  $f$ .

4) Find the minimum value of the surface area of a rectangular box that has a volume of 1000 unit<sup>3</sup>.

**Note:** If  $x, y$  and  $z$  represent the lengths of the three sides of a rectangular box, then the surface area and volume are given by  $2(xy + yz + zx)$  and  $xyz$ , respectively.

600

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 600

**1 point**

5) Consider a function  $f(x, y, z) = x^3 + 3y^3 + z^3 - 4x - 6y - 4z$ . If  $A$  is the the Hessian matrix of  $f(x, y, z)$  at  $(0, 1, 1)$ , then which of the following option(s) is(are) true?

**1 point**

☒  $\det(A) = 0$

☒  $\text{Rank}(A) = 2$

☒  $\text{Nullity}(A) = 1$

☐  $\det(A) = 648$

☐  $\text{Rank}(A) = 3$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$\det(A) = 0$$

$$\text{Rank}(A) = 2$$

$$\text{Nullity}(A) = 1$$

6) Consider the following function

**1 point**

$$f(x, y) = e^x + e^y - e^{x+y}$$

Which of the following option(s) is (are) true?

- ☐ Hessian matrix of  $f$  at point  $(1, 1)$  is  $\begin{bmatrix} 1-e & -e^2 \\ -e^2 & 1-e \end{bmatrix}$ .
- ☒ Hessian matrix of  $f$  at point  $(1, 1)$  is  $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$ .
- ☐  $f$  has a point of maximum.
- ☒  $f$  has a saddle point.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Hessian matrix of  $f$  at point  $(1, 1)$  is  $\begin{bmatrix} e-e^2 & -e^2 \\ -e^2 & e-e^2 \end{bmatrix}$ .

$f$  has a saddle point.

7) Choose the correct statements.

**1 point**

- ☐ If  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  then the Hessian matrix  $H_f(x, y, z)$  is same for all  $(x, y, z) \in \mathbb{R}^3$ .
- ☒ If  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  is a linear transformation then the Hessian matrix  $H_f(x, y, z)$  is same for all  $(x, y, z) \in \mathbb{R}^3$ .
- ☒ For each  $n \in \mathbb{N}$ , there exists a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  such that the rank of the Hessian matrix  $H_f(x_1, x_2, \dots, x_n)$  is  $n$  for some  $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ .
- ☒

For each  $n \in \mathbb{N}$ , there exists a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  such that the Hessian matrix  $H_f(x_1, x_2, \dots, x_n)$  is symmetric for all  $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

If  $f : \mathbb{R}^3 \rightarrow \mathbb{R}$  is a linear transformation then the Hessian matrix  $H_f(x, y, z)$  is same for all  $(x, y, z) \in \mathbb{R}^3$ .

For each  $n \in \mathbb{N}$ , there exists a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  such that the rank of the Hessian matrix  $H_f(x_1, x_2, \dots, x_n)$  is  $n$  for some  $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ .

For each  $n \in \mathbb{N}$ , there exists a map  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  such that the Hessian matrix  $H_f(x_1, x_2, \dots, x_n)$  is symmetric for all  $(x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ .

8) The maximum value of the function  $-2x^2 + 2xy - y^2 + 12x - 4y - 7$  is  
13

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 13

**1 point**

9) Define a function

**1 point**

$$f(x, y) = \begin{cases} (x - y)^2 \sin\left(\frac{1}{x - y}\right) & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}$$

Choose the set of correct options.

- ☒  $f$  is continuous at  $(0, 0)$ .
- ☐  $f_x(0, 0) = f_y(0, 0)$
- ☒  $f_x(0, 0) = f_y(0, 0)$
- ☒  $f$  is differentiable at  $(0, 0)$ .

Yes, the answer is correct.

Score: 1

**Accepted Answers:** $f$  is continuous at  $(0, 0)$ .

$$f_x(0, 0) = f_y(0, 0)$$

 $f$  is differentiable at  $(0, 0)$ .

10) A company produces two types of chocolates, one that sells for Rs.3 and another at Rs.2. The total revenue, in lakhs, from the sale of  $x$  lakhs chocolates at Rs.2 each and  $y$  lakhs at Rs. 3 each is given by

$$F(x, y) = 2x + 3y.$$

The total cost in lakhs for producing  $x$  lakhs of the Rs.2 chocolates and  $y$  lakhs of the Rs.3 chocolates is given by

$$G(x, y) = 2x^2 - 2xy + y^2 - 9x + 6y + 7.$$

Then the maximum profit in lakhs is

[Enter your answer upto two decimal places]

11.25

Yes, the answer is correct.

Score: 1

**Accepted Answers:**

(Type: Numeric) 11.25

**1 point**

11) (MSQ) Let  $f : \mathbb{R}^2 \rightarrow \mathbb{R}$  be a multivariable function and  $(x_0, y_0) \in \mathbb{R}^2$ . Consider the following statements: **1 point**

P:  $f$  is continuous at  $(x_0, y_0)$ .

Q: The partial derivatives of  $f$  exist and are continuous in a neighborhood of  $(x_0, y_0)$ .

R: The tangent plane to the graph of  $f$  exists at  $(x_0, y_0)$ .

Choose the correct statements from the following.

- ☐ P implies that  $f$  is differentiable at  $(x_0, y_0)$ .
- ☒ If  $f$  is differentiable at  $(x_0, y_0)$ , then P is satisfied.
- ☒ Q implies that  $f$  is differentiable at  $(x_0, y_0)$ .
- ☐ If  $f$  is differentiable at  $(x_0, y_0)$ , then Q is satisfied.
- ☒ R implies that  $f$  is differentiable at  $(x_0, y_0)$ .
- ☒ If  $f$  is differentiable at  $(x_0, y_0)$ , then R is satisfied.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If  $f$  is differentiable at  $(x_0, y_0)$ , then P is satisfied.

Q implies that  $f$  is differentiable at  $(x_0, y_0)$ .

R implies that  $f$  is differentiable at  $(x_0, y_0)$ .

If  $f$  is differentiable at  $(x_0, y_0)$ , then R is satisfied.

12) (NAT) A particle moving in a potential field tends to settle at points where the potential energy is a local minimum (stable equilibrium point). Consider a particle moving in  $\mathbb{R}^2$  where the potential energy as a function of position is given by

$$V(x, y) = x^4 + y^4 - 4xy + 1,$$

for all  $(x, y) \in \mathbb{R}^2$ . Using the Hessian test, find the potential energy at stable equilibrium points.

-1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -1

**1 point**